

AWS Architecture Services

Frontend

The frontend refers to the client-side of the application, focusing on user experience and interface. It's what users interact with directly more like the face of the app.

Route 53: Domain name system (DNS) management for routing users to the application, providing high availability and scalability. Think of it like a phonebook of internet. It helps users find your app by translating domain names into IP addresses.

CloudFront: Content delivery network (CDN) for distributing static assets, reducing latency, and improving user experience. Imagine a network of caching servers around the world, storing copies of your website's static files, so users get faster access no matter the location.



Amplify Hosting: Managed hosting for web applications, providing automated deployment, SSL/TLS, and integration with other AWS services. It's like having a hosting service that takes care of all the technical details, so you can focus on building your app.

Backend

The backend refers to the server-side of the application, handling business logic, database interactions, and API integrations. Basically this is where the magic happens.

App Runner: Containerized application hosting, simplifying deployment and scaling of containerized applications. It's more of like having a managed platform for running your containerized apps, without worrying about the underlying infrastructure.

Lambda: Serverless compute for handling small, event-driven tasks, reducing operational overhead. Think of it like a service that runs your code only when needed, without requiring server management.



Storage

Storage refers to the services used to store and manage data.

S3 (3D assets + static files): Object storage for hosting static files, providing durability, scalability, and high availability. It's like having a massive, secure hard drive in the cloud, where you can store and serve files.

Databases

Databases are specialized storage services for structured data.

DynamoDB (catalog): NoSQL database for handling high-traffic catalog data, providing low latency and scalability. It's designed for fast, efficient data retrieval and storage.

RDS PostgreSQL (orders / users): Relational



database for managing orders and user data, providing ACID compliance and SQL querying capabilities. It's a managed database service, handling the underlying infrastructure, so you can focus on your data.

Performance

Performance services focus on optimizing application speed and efficiency.

ElastiCache Redis: In-memory caching for improving application performance, reducing database load, and enhancing user experience. It's like having a super-fast cache layer, storing frequently accessed data.

SQS + Lambda workers: Message queuing and processing for handling asynchronous tasks, decoupling application components, and improving scalability. It's like having a message broker, handling task queues and processing, so your app can focus on other things.



Security

Security services protect the application and its data from threats and vulnerabilities.

WAF: Web application firewall for protecting against common web exploits and vulnerabilities. It's like having a security guard, monitoring and blocking malicious traffic.

Shield: DDoS protection for safeguarding against distributed denial-of-service attacks. It's like having a shield, protecting your app from overwhelming traffic.

KMS: Key management service for encrypting sensitive data and managing encryption keys. It's like having a secure key vault, protecting your encryption keys.

Secrets Manage: Secrets management for securely storing and retrieving sensitive information, such as database credentials. It's like having a secure password manager, storing and retrieving secrets.



Monitoring

Monitoring services track application performance, errors, and security issues.

CloudWatch: Monitoring and logging service for tracking application performance, errors, and security issues. It's like having a dashboard, showing real-time insights into your app's health.

Cost Explorer: Cost management tool for analyzing and optimizing AWS costs, ensuring efficient resource utilization. It's like having a financial advisor, helping you optimize your AWS spending.

